

Sîrghie Matei-Ștefan

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EDUCATION

University of Bucharest Oct. 2024 - Jul. 2028
BEng Computer Science *Current GPA: 9.56/10.0*
Tudor Vianu National High School of Computer Science Sept. 2020 - Jul. 2024
Computer Science specialization *GPA: 9.41/10.0*
Courses:
Object-Oriented Programming, Data Structures & Algorithms, Numerical methods, Algebra and geometry, Calculus, Physics

EXPERIENCE

University of Bucharest, Faculty of Mathematics and Informatics | *Student Tutor* Oct. 2025 - Present

- Approved by faculty to instruct students in Computer Programming 1 (C) and 2 (C++) for practical exam credit.
- Guided undergraduate students through fundamental syntax, algorithmic design, and data structures.
- Reviewed and debugged student code, providing constructive feedback on logic, memory management, and overall execution efficiency.

Asociația Studenților la Matematică și Informatică | *Design and PR member* Nov. 2024 – Present

- Managed digital asset workflows using DaVinci Resolve, CapCut, Krita, and Photoshop.
- Assisted the technical group with website management, contributing to the user interface (UI).
- Designed PR materials in Canva and analyzed TikTok engagement metrics to iterate on PR strategies(UX).

PROJECTS

Song-Approximation | *Python, Audio Processing, STFT, SVD* Jun. 2025

- Created a lossy .wav compression algorithm translating time-domain signals via Short-Time Fourier Transform.
- Reduced matrix dimensionality and storage size using Singular Value Decomposition (SVD) and QR factorization.
- Reconstructed audio via Inverse STFT, calculating cumulative errors.

Find Rakis (Space Simulation) | *C++23, SFML, CMake, OOP, CI/CD* Oct. 2025

- Developed a 2D C++ simulation with custom OOP architecture to handle logic, resources.
- Implemented Perlin noise and dithering and used Integrated GitHub Actions CI/CD for automation.
- Simulated complex 2D space physics, managing gravitational wells, momentum.

ChessHacker (Real-Time Analyzer) | *Python, Playwright, Stockfish, DOM Parsing* Mar. 2025

- Created an app utilizing Playwright to dynamically parse live HTML board and evaluate position.
- Integrated the Stockfish engine with adjustable depth levels to analyze parsed positions.

3D Hyperboloid Visualizer | *JavaScript, Three.js, Node.js, HTML5 Canvas* Mar. 2026

- Developed a real-time 3D rendering web application utilizing Three.js to dynamically animate a hyperboloid.
- Implemented parameter adjustments, allowing real-time translation and transformation based on user input.

AWARDS

Romanian National Olympiad in Informatics: Bronze Medal (30th, 2017) | Finalist (58th in 2024, 84th in 2022, 55th in 2019)
Aerospace Competitions: Achieved a Top 10 Placement in the Qube2Space International Competition

SKILLS

Programming Languages: C/C++(8+ years of Competitive Programming), C#, Python, JavaScript, Assembly, SQL
Frameworks & Libraries: Node.js(Multiple Math Visualization websites), Express, Playwright, SFML, Tailwind
Game & Simulation Engines: Godot Engine, Unity Engine(Multiple unfinished Unity Game Projects)
Developer Tools & CI/CD: Git/GitHub, CMake, Unix/Linux Shell, GitHub Actions, Docker, MATLAB, AutoCAD
Creative & Media Tools: DaVinci Resolve, Adobe Photoshop, Krita, CapCut, Canva, Figma
Spoken Languages: Romanian (Native), English (C2 Cambridge)